# Objectives

1. Spark SQL Dcumentation
2. Create database demo
3. Data tab in the UI
4. SHOW command
5. DESCRIBE command
6. Find the current database

```
create database IF not exists demo;
```

OK

```
SHOW DATABASES;
```

Table

| | databaseName |
|---|---|
| 1 | default |
| 2 | demo |

2 rows

```
DESC DATABASE demo;
```

Table

| | database_description_item | database_description_value |
|---|---|---|
| 1 | Catalog Name | spark_catalog |
| 2 | Namespace Name | demo |
| 3 | Comment | |
| 4 | Location | dbfs:/user/hive/warehouse/demo.db |
| 5 | Owner | root |

5 rows

```
DESC DATABASE EXTENDED demo;
```

Table

| | database_description_item | database_description_value |
|---|---|---|

| | | |
|---|---|---|
| 1 | Catalog Name | spark_catalog |
| 2 | Namespace Name | demo |
| 3 | Comment | |
| 4 | Location | dbfs:/user/hive/warehouse/demo.db |
| 5 | Owner | root |
| 6 | Properties | |

6 rows

## get current database name

```
SELECT CURRENT_DATABASE();
```

Table

| | current_schema() |
|---|---|
| 1 | default |

1 row

### get tables present in nhe current database

```
show tables;
```

Query returned no results

### Change database as required

```
USE demo;
```

OK

```
SELECT CURRENT_DATABASE();
```

Table

| | current_schema() |
|---|---|
| 1 | demo |

1 row

## Create Managed Table

```
%run "../includes/configuration"
```

```
%python
race_results_df = spark.read.parquet(f"{presentation_folder_path}/race_results")
```

```
%python
race_results_df.write.mode("overwrite").format("parquet").saveAsTable("demo.race_results_python")
```

```
SHOW TABLES;
```

Table

| | database | tableName | isTemporary |
|---|---|---|---|
| 1 | demo | race_results_python | false |
| 2 | demo | race_results_sql | false |

2 rows

```
DESC  race_results_python
```

Table

| | col_name | data_type | comment |
|---|---|---|---|
| 1 | race_year | int | null |
| 2 | race_name | string | null |
| 3 | race_date | timestamp | null |
| 4 | circuit_location | string | null |
| 5 | driver_name | string | null |
| 6 | driver_nationality | string | null |
| 7 | team | string | null |
| 8 | grid | int | null |
| 9 | fastest_lap | int | null |
| 10 | race_time | string | null |
| 11 | points | float | null |
| 12 | position | int | null |
| 13 | created_date | timestamp | null |

13 rows

```
DESC EXTENDED race_results_python
```

Table

| | col_name | data_type | comment |
|---|---|---|---|
| 1 | race_year | int | null |

| | | | |
|---|---|---|---|
| 2 | race_name | string | null |
| 3 | race_date | timestamp | null |
| 4 | circuit_location | string | null |
| 5 | driver_name | string | null |
| 6 | driver_nationality | string | null |
| 7 | team | string | null |
| 8 | grid | int | null |
| 9 | fastest_lap | int | null |
| 10 | race_time | string | null |
| 11 | points | float | null |
| 12 | position | int | null |
| 13 | created_date | timestamp | null |
| 14 | | | |
| 15 | # Detailed Table Information | | |

28 rows

```
SELECT *
FROM demo.race_results_python
where race_year = 2020;
```

Table

| | race_year | race_name | race_date | circuit_location | driver_name | driver_nationality |
|---|---|---|---|---|---|---|
| 1 | 2020 | Bahrain Grand Prix | 2020-11-29T14:10:00.000+00:... | Sakhir | Lewis Hamilton | British |
| 2 | 2020 | Bahrain Grand Prix | 2020-11-29T14:10:00.000+00:... | Sakhir | Max Verstappen | Dutch |
| 3 | 2020 | Bahrain Grand Prix | 2020-11-29T14:10:00.000+00:... | Sakhir | Alexander Albon | Thai |
| 4 | 2020 | Bahrain Grand Prix | 2020-11-29T14:10:00.000+00:... | Sakhir | Lando Norris | British |
| 5 | 2020 | Bahrain Grand Prix | 2020-11-29T14:10:00.000+00:... | Sakhir | Carlos Sainz | Spanish |
| 6 | 2020 | Bahrain Grand Prix | 2020-11-29T14:10:00.000+00:... | Sakhir | Pierre Gasly | French |
| 7 | 2020 | Bahrain Grand Prix | 2020-11-29T14:10:00.000+00:... | Sakhir | Daniel Ricciardo | Australian |
| 8 | 2020 | Bahrain Grand Prix | 2020-11-29T14:10:00.000+00:... | Sakhir | Valtteri Bottas | Finnish |
| 9 | 2020 | Bahrain Grand Prix | 2020-11-29T14:10:00.000+00:... | Sakhir | Esteban Ocon | French |
| 10 | 2020 | Bahrain Grand Prix | 2020-11-29T14:10:00.000+00:... | Sakhir | Charles Leclerc | Monegasque |
| 11 | 2020 | Bahrain Grand Prix | 2020-11-29T14:10:00.000+00:... | Sakhir | Daniil Kvyat | Russian |
| 12 | 2020 | Bahrain Grand Prix | 2020-11-29T14:10:00.000+00:... | Sakhir | George Russell | British |
| 13 | 2020 | Bahrain Grand Prix | 2020-11-29T14:10:00.000+00:... | Sakhir | Sebastian Vettel | German |
| 14 | 2020 | Bahrain Grand Prix | 2020-11-29T14:10:00.000+00:... | Sakhir | Nicholas Latifi | Canadian |

340 rows

```
CREATE or replace TABLE demo.race_results_sql
AS
SELECT *
FROM demo.race_results_python
where race_year = 2020;
```

Query returned no results

```
DESC EXTENDED demo.race_results_sql;
```

Table

| | col_name | data_type |
|---|---|---|

| | | |
|---|---|---|
| 1 | race_year | int |
| 2 | race_name | string |
| 3 | race_date | timestamp |
| 4 | circuit_location | string |
| 5 | driver_name | string |
| 6 | driver_nationality | string |
| 7 | team | string |
| 8 | grid | int |
| 9 | fastest_lap | int |
| 10 | race_time | string |
| 11 | points | float |
| 12 | position | int |
| 13 | created_date | timestamp |
| 14 | | |

32 rows

# External Table

## Objective

1. Create External Table using Python
2. Create External Table using SQL
3. Effect of dropping an External Table

### using python

```
%python
race_results_df.write.mode("overwrite").format("parquet").option("path", f"{presentation_folder_path}/race_results_ext_py").saveAsT
```

```
desc extended demo.race_results_ext_py;
```

Table

| | col_name | data_type | comment |
|---|---|---|---|
| 1 | race_year | int | null |
| 2 | race_name | string | null |
| 3 | race_date | timestamp | null |
| 4 | circuit_location | string | null |
| 5 | driver_name | string | null |
| 6 | driver_nationality | string | null |
| 7 | team | string | null |
| 8 | grid | int | null |
| 9 | fastest_lap | int | null |
| 10 | race_time | string | null |
| 11 | points | float | null |
| 12 | position | int | null |
| 13 | created_date | timestamp | null |
| 14 | | | |
| 15 | # Detailed Table Information | | |

28 rows

### Creating External Table using SQL

```
CREATE TABLE demo.race_results_ext_sql
(
race_year INT,
race_name STRING,
race_date TIMESTAMP,
circuit_location STRING,
driver_name STRING,
driver_nationality STRING,
team STRING,
grid INT,
fastest_lap INT,
race_time STRING,
points FLOAT,
position INT,
created_date TIMESTAMP
)
USING parquet
LOCATION "/mnt/formula1dl1204/presentation/race_results_ext_sql";
```

OK

```
drop table demo.race_results_ext_sql;
```

OK

```
show tables in demo;
```

Table

| | database | tableName | isTemporary |
|---|---|---|---|
| 1 | demo | race_results_ext_py | false |
| 2 | demo | race_results_ext_sql | false |
| 3 | demo | race_results_python | false |
| 4 | demo | race_results_sql | false |

4 rows

```
insert into demo.race_results_ext_sql
select * from demo.race_results_ext_py where race_year = 2020;
```

OK

```
select * from demo.race_results_ext_sql;
```

Table

| | race_year | race_name | race_date | circuit_location | driver_name | driver_nationality |
|---|---|---|---|---|---|---|
| 1 | 2020 | Abu Dhabi Grand P... | 2020-12-13T13:10:00.000+00:... | Abu Dhabi | Max Verstappen | Dutch |
| 2 | 2020 | Abu Dhabi Grand P... | 2020-12-13T13:10:00.000+00:... | Abu Dhabi | Valtteri Bottas | Finnish |
| 3 | 2020 | Abu Dhabi Grand P... | 2020-12-13T13:10:00.000+00:... | Abu Dhabi | Lewis Hamilton | British |
| 4 | 2020 | Abu Dhabi Grand P... | 2020-12-13T13:10:00.000+00:... | Abu Dhabi | Alexander Albon | Thai |

| | | | | | | |
|---|---|---|---|---|---|---|
| 5 | 2020 | Abu Dhabi Grand P... | 2020-12-13T13:10:00.000+00:... | Abu Dhabi | Lando Norris | British |
| 6 | 2020 | Abu Dhabi Grand P... | 2020-12-13T13:10:00.000+00:... | Abu Dhabi | Carlos Sainz | Spanish |
| 7 | 2020 | Abu Dhabi Grand P... | 2020-12-13T13:10:00.000+00:... | Abu Dhabi | Daniel Ricciardo | Australian |
| 8 | 2020 | Abu Dhabi Grand P... | 2020-12-13T13:10:00.000+00:... | Abu Dhabi | Pierre Gasly | French |
| 9 | 2020 | Abu Dhabi Grand P... | 2020-12-13T13:10:00.000+00:... | Abu Dhabi | Esteban Ocon | French |
| 10 | 2020 | Abu Dhabi Grand P... | 2020-12-13T13:10:00.000+00:... | Abu Dhabi | Lance Stroll | Canadian |
| 11 | 2020 | Abu Dhabi Grand P... | 2020-12-13T13:10:00.000+00:... | Abu Dhabi | Daniil Kvyat | Russian |
| 12 | 2020 | Abu Dhabi Grand P... | 2020-12-13T13:10:00.000+00:... | Abu Dhabi | Kimi Räikkönen | Finnish |
| 13 | 2020 | Abu Dhabi Grand P... | 2020-12-13T13:10:00.000+00:... | Abu Dhabi | Charles Leclerc | Monegasque |
| 14 | 2020 | Abu Dhabi Grand P... | 2020-12-13T13:10:00.000+00:... | Abu Dhabi | Sebastian Vettel | German |

340 rows

```
select * from demo.race_results_ext_py;
```

Table

| | race_year | race_name | race_date | circuit_location | driver_name | driver_nationality |
|---|---|---|---|---|---|---|
| 1 | 2009 | Spanish Grand Prix | 2009-05-10T12:00:00.000+00:... | Montmeló | Jenson Button | British |
| 2 | 2009 | Spanish Grand Prix | 2009-05-10T12:00:00.000+00:... | Montmeló | Rubens Barrichello | Brazilian |
| 3 | 2009 | Spanish Grand Prix | 2009-05-10T12:00:00.000+00:... | Montmeló | Mark Webber | Australian |
| 4 | 2009 | Spanish Grand Prix | 2009-05-10T12:00:00.000+00:... | Montmeló | Sebastian Vettel | German |
| 5 | 2009 | Spanish Grand Prix | 2009-05-10T12:00:00.000+00:... | Montmeló | Fernando Alonso | Spanish |
| 6 | 2009 | Spanish Grand Prix | 2009-05-10T12:00:00.000+00:... | Montmeló | Felipe Massa | Brazilian |
| 7 | 2009 | Spanish Grand Prix | 2009-05-10T12:00:00.000+00:... | Montmeló | Nick Heidfeld | German |
| 8 | 2009 | Spanish Grand Prix | 2009-05-10T12:00:00.000+00:... | Montmeló | Nico Rosberg | German |
| 9 | 2009 | Spanish Grand Prix | 2009-05-10T12:00:00.000+00:... | Montmeló | Lewis Hamilton | British |
| 10 | 2009 | Spanish Grand Prix | 2009-05-10T12:00:00.000+00:... | Montmeló | Timo Glock | German |
| 11 | 2009 | Spanish Grand Prix | 2009-05-10T12:00:00.000+00:... | Montmeló | Robert Kubica | Polish |
| 12 | 2009 | Spanish Grand Prix | 2009-05-10T12:00:00.000+00:... | Montmeló | Nelson Piquet Jr. | Brazilian |
| 13 | 2009 | Spanish Grand Prix | 2009-05-10T12:00:00.000+00:... | Montmeló | Kazuki Nakajima | Japanese |
| 14 | 2009 | Spanish Grand Prix | 2009-05-10T12:00:00.000+00:... | Montmeló | Giancarlo Fisichella | Italian |

10,000+ rows | Truncated data due to row limit

# Views

## Objective

1. Create Temp View
2. Create Global Temp View
3. Create Permamnent View

```
create or replace temporary view v_race_results
as
select * from demo.race_results_python;
```

OK

```
create or replace global temporary view gv_race_results
as
select * from demo.race_results_python;
```

OK

```
select * from global_temp.gv_race_results;
```

Table

| | race_year | race_name | race_date | circuit_location | driver_name | driver_nationality |
|---|---|---|---|---|---|---|
| 1 | 2009 | Spanish Grand Prix | 2009-05-10T12:00:00.000+00:... | Montmeló | Jenson Button | British |
| 2 | 2009 | Spanish Grand Prix | 2009-05-10T12:00:00.000+00:... | Montmeló | Rubens Barrichello | Brazilian |
| 3 | 2009 | Spanish Grand Prix | 2009-05-10T12:00:00.000+00:... | Montmeló | Mark Webber | Australian |
| 4 | 2009 | Spanish Grand Prix | 2009-05-10T12:00:00.000+00:... | Montmeló | Sebastian Vettel | German |
| 5 | 2009 | Spanish Grand Prix | 2009-05-10T12:00:00.000+00:... | Montmeló | Fernando Alonso | Spanish |
| 6 | 2009 | Spanish Grand Prix | 2009-05-10T12:00:00.000+00:... | Montmeló | Felipe Massa | Brazilian |
| 7 | 2009 | Spanish Grand Prix | 2009-05-10T12:00:00.000+00:... | Montmeló | Nick Heidfeld | German |
| 8 | 2009 | Spanish Grand Prix | 2009-05-10T12:00:00.000+00:... | Montmeló | Nico Rosberg | German |
| 9 | 2009 | Spanish Grand Prix | 2009-05-10T12:00:00.000+00:... | Montmeló | Lewis Hamilton | British |
| 10 | 2009 | Spanish Grand Prix | 2009-05-10T12:00:00.000+00:... | Montmeló | Timo Glock | German |
| 11 | 2009 | Spanish Grand Prix | 2009-05-10T12:00:00.000+00:... | Montmeló | Robert Kubica | Polish |
| 12 | 2009 | Spanish Grand Prix | 2009-05-10T12:00:00.000+00:... | Montmeló | Nelson Piquet Jr. | Brazilian |
| 13 | 2009 | Spanish Grand Prix | 2009-05-10T12:00:00.000+00:... | Montmeló | Kazuki Nakajima | Japanese |
| 14 | 2009 | Spanish Grand Prix | 2009-05-10T12:00:00.000+00:... | Montmeló | Giancarlo Fisichella | Italian |

10,000+ rows | Truncated data due to row limit

```
create or replace view demo.pv_race_results
as
select * from demo.race_results_python;
```

OK

```
show tables;
```

Table

| | database | tableName | isTemporary |
|---|---|---|---|
| 1 | demo | pv_race_results | false |
| 2 | demo | race_results_ext_py | false |
| 3 | demo | race_results_ext_sql | false |
| 4 | demo | race_results_python | false |
| 5 | demo | race_results_sql | false |
| 6 | | v_race_results | true |

6 rows